GridSense AI - Release Note

built 2026-11-06 07:45 UTC - figures fetched from the live deployment and from git at build time

1. Where it stands

Frontend	https://grid-senseai.vercel.app -> HTTP 200
Backend	https://grid-senseai-backend.fastapicloud.dev -> HTTP 200 (health), 200 (validation report)
Model trained on	51,959 hourly rows - 3 zones over 722 days (entsoe 51,959); weather nasa_power 51,689, unavailable 270
API serves from	1,008 in-memory rows - 336 h per zone, the tail kept by ROWS_PER_ZONE_BUFFER = 336
Release window	commits on main dated 2026-11-05 or later
Backend delta	7 commits; 28 files changed, 2257 insertions(+), 518 deletions(-)
Frontend delta	14 commits; 36 files changed, 3838 insertions(+), 779 deletions(-)
Overall verdict	WARN across 13 release gates: 9 PASS, 4 WARN, 0 FAIL

The product is deployed and answers with real data. What follows is deliberately split into what shipped, what broke and was repaired, what is still open, and what the next phase needs - with the per-feed currency table first, because on a grid dashboard 'current' has to mean something measurable.

2. Current condition, per feed

Feed	Source	Rows (served / trained)	Latest hour	Age now	What that means
Load	ENTSO-E transparent day-ahead	1,008 / 51,959	2026-11-04T17:00	38.8 h	real measured hours, but a frozen snapshot: the API serves the committed frame and advances it by rebuild plus redeploy, so this ages with the load
Weather	NASA POWER hourly	738 / 51,689	2026-10-31T23:00	128.8 h	lags load by 3.8 days; 270 served rows have no weather at all, so every live forecast runs on the fallback
Capacity	p99-of-load proxy	derived per zone	n/a	n/a	a distribution peak observed in the data, not an operator's rated capacity
Serving window	backend scheduler	3 zones	2026-11-04T17:00	38.8 h	ok

Served rows count the in-memory tail the API answers from; trained rows count the full feature table the models were fitted on. They are different scopes, not different measurements of the same thing.

Which model is answering right now, per zone:

Zone	Model answering	Horizon	Hour +1 median	Origin age
Spain	no / weather / fallback	24 h	33,219 MW	38.8 h
Greece	no / weather / fallback	24 h	6,202 MW	38.8 h
Italy South	no / weather / fallback	24 h	2,677 MW	38.8 h

All three zones are answered by the reduced-feature fallback because NASA POWER is 90 hours behind load. The weather-aware model is trained and served, and answers as soon as weather catches up - it is not dead code, it is starved.

3. What was added

Backend - 7 commits (3 of them capability changes):

- 0128745 feat(api): serve the fallback profile and declare one horizon per feed in /health
- 682615c feat(models): retrain on the real window and add a calibrated no-weather fallback
- 9c6a48e feat(data): rebuild the feature frame on two years of real ENTSO-E and NASA POWER hours

Plus 2 documentation, test or build commits in the same window.

Frontend - 14 commits (8 of them capability changes):

- e2c3d38 feat(ui): measured colophon footer with provenance and rights
- 2615d94 feat(ui): show the scale the risk score is measured against
- 14b1dfb feat(ui): say what the product does, next to the wordmark
- a7a9284 feat(ui): give the phone first screen something worth reading
- ea21c69 feat(ui): a phone layout that reorders the dashboard instead of shrinking it
- b0b0190 feat(3d): radar sweep with radial falloff, selection reticle and honest frame timing
- 8a53e54 feat(ui): one stroked SVG icon family, loading skeletons and real favicons
- 3b73f80 feat(ui): signal integrity rail stating what each feed actually reaches

Plus 1 documentation, test or build commits in the same window.

4. What broke, and what fixed it

7 commits in the window are fixes. Each one below was found by running the system and measuring the result, not by reading the code.

- fcf0744 fix(validation): publish repo-relative artifact paths on a public endpoint - /validation echoed str(MODELS_DIR / name), which exposed the absolute
- 43fa10d fix(models): refuse to forecast weather-needing models from a weather-less frame - The no-weather path fills absent weather columns with zeros, which is only sound while
- b44f979 fix(3d): stop warning about the shadow type three already substitutes - three 0.185 logs PCFSoftShadowMap as deprecated and silently uses PCFShadowMap, so selecting PCF explicitly changes nothing in the render but clears t
- 028aa84 fix(ui): show release gates as a map, not a 3px hairline - The summary bar was measured at 3 CSS pixels tall, which reads as a divider rather than a result, so 9 PASS / 4 WARN looked like missing content. One
- e524e3c fix(chart): restore the phone forecast plot and the median line - The 24h plot rendered as a ~470px void below lg. The chart box was a
- 8b77cbf fix(copy): say what each feed is, not just "proxy" - Bare "PROXY" read as an apology for the data. The load and weather feeds
- 2deb39c fix(ui): wrap instead of clip, hit 32px targets, and label what colour alone cannot - Measured against the real feed at 375, 768 and 1440 px, the panel pass removed the

Two of the measured defects are worth calling out because a screenshot would not have caught either of them:

- The phone forecast panel was a ~470px void. A flex item sized by min-height clamping does not have a definite height, so the chart container's height:100% resolved to 0. Measured: container 331x0, no SVG surface. Now 331x280 with a drawn surface.
- The P50 median line had never rendered at any width. Recharts' AreaChart renders only Area children and silently drops Line children - no warning, no error, no console output. Switched to ComposedChart; measured 3 line paths, median stroke #35d6e8, dasharray none.

5. What remains open

The release gates that are WARN, not PASS (4 of 13):

- weekly coverage spread <=0.15 (band not decaying across the window)
- >=80% of real overload events raise an alarm while running
- >=50% of events forecast-warned >=3h before onset
- median STABLE forecast lead before onset >=3h (unbroken warning from h+1)

Beyond the gates, five things are genuinely unfinished:

- No data refresh path. The API loads the committed feature frame at startup and keeps the last 336 h per zone; there is no request-time or scheduled fetch, so the newest hour ages one hour per hour after each deploy. Measured: frame shipped 9c6a48e at 2026-11-05 23:52+0500, when the newest load hour was 25.9 h old - it is now 38.8 h old, 12.9 h later. The UI reports these ages truthfully; the fix is a scheduled rebuild, not new copy.
- Rated capacity. Every MWa figure is a p99-of-load proxy derived from the same load series it scores, so utilisation percentages are self-referential. This is stated in the UI, the model card and the audit report; it is not fixed.
- Weather currency. NASA POWER arrives on a multi-day lag, so the deployed product runs on the fallback model. The weather-aware path is therefore exercised in replay, not in production serving.
- Documentation. 2 superseded-figure patterns still match in docs/ ("13/13" in 1 file(s); "synthetic" in 10 file(s)). README, model card and this note are synced to the retrain; the numbered docs are not fully.

- Domain. Load is ENTSO-E (Spain, Greece, Italy South) and weather is NASA POWER. There is no public Pakistani hourly feed to train on, so NTDC/DISCO is stood in for and no accuracy figure here transfers to it without retraining.

6. Future direction

Ordered by what each step would actually change about the numbers on screen.

- Rated capacity per zone first. It is the cheapest high-impact change: it converts utilisation and risk from a self-referential scale into one an operator can act on. Needs a published transmission capacity figure per bidding zone, or TSO network development plan data.
- A scheduled refresh of the served frame. Either a job that re-pulls ENTSO-E and NASA POWER, rebuilds the parquet and redeploys, or a request-time fetch behind a cache. Deferring it is a trade, not an oversight: with the frame committed, the runtime holds no third-party credentials and cannot be failed by an upstream outage, which is the behaviour a demo needs.
- A forward-looking weather feed with a horizon that matches the forecast horizon. One candidate was evaluated and rejected on a measured variable mismatch against the trained feature set; adopting any such feed means retraining, and the closed-loop replay is already in place to show whether the swap helps.
- Close the two WARN gates rather than accept them, then re-run the negative control so a passing board still means something.
- Re-point the same pipeline at a non-European zone to prove it generalises: the feature builder, model set and replay engine are zone-parameterised, so the second domain is a data task, not a rewrite.
- Only then consider real-time alerts or a dispatch-facing surface. Nothing in the current evidence supports turning a research prototype into an operational instruction, and the UI says so on every page.

7. How to re-check this note

```
curl -s https://grid-senseai-backend.fastapicloud.dev/health | python -m json.tool | head -40
curl -s https://grid-senseai-backend.fastapicloud.dev/validation | python -c "import json,sys; d=json.load(sys.stdin); print(d['overall'], [g['result'] for g in d['gates']])"
python -c "import pandas as pd; f=pd.read_parquet('data/features/training_features.parquet'); print(len(f), f['zone'].nunique())"   # trained rows / zones
git -C gridsense-backend log --since='2026-11-05 00:00' --oneline
git -C gridsense-frontend log --since='2026-11-05 00:00' --oneline
python scripts/build_release_note_pdf.py   # rebuilds this file
```

Sources: backend https://github.com/huzaifahimad/Grid_Sense.AI-Backend- ; frontend https://github.com/huzaifahimad/Grid_Sense.AI-frontend- ; built by build_release_note_pdf.py on 2026-11-06 07:45 UTC.